

Determine the limits between which the temperature fluctuates when the motor operates on a load cycle consisting of alternate period of 1 hr. on full-load and 1 hr. on no-load, steady state conditions having been established. [28.7° C, 21.3° C]

21. A motor with a thermal time constant of 45 min. has a final temperature rise of 75°C on continuous rating (a) What is the temperature rise after one hour at this load? (b) If the temperature rise on one-hour rating is 75°C, find the maximum steady temperature at this rating (c) When working at its one-hour rating, how long does it take the temperature to increase from 60°C to 75°C? [(a) 55° C (b) 102° C (c) 20 min]

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OBJECTIVE TESTS – 30

- The speed of a d.c. motor can be controlled by varying
 - its flux per pole
 - resistance of armature circuit
 - applied voltage
 - all of the above
- The most efficient method of increasing the speed of a 3.75 kW d.c. shunt motor would be themethod.
 - armature control
 - flux control
 - Ward-Leonard
 - tapped-field control
- Regarding Ward-Leonard system of speed control which statement is **false**?
 - It is usually used where wide and very sensitive speed control is required.
 - It is used for motors having ratings from 750 kW to 4000 kW
 - Capital outlay involved in the system is right since it uses two extra machines.
 - It gives a speed range of 10 : 1 but in one direction only.
 - It has low overall efficiency especially at light loads.
- In the rheostatic method of speed control for a d.c. shunt motor, use of armature diverter makes the method
 - less wasteful
 - less expensive
 - unsuitable for changing loads
 - suitable for rapidly changing loads
- The chief advantage of Ward-Leonard system of d.c. motor speed control is that it
 - can be used even for small motors
 - has high overall efficiency at all speeds
 - gives smooth, sensitive and wide speed control
 - uses a flywheel to reduce fluctuations in power demand
- The flux control method using paralleling of field coils when applied to a 4-pole series d.c. motor can give speeds.
 - 2
 - 3
 - 4
 - 6
- The series-parallel system of speed control of series motors widely used in traction work gives a speed range of about
 - 1 : 2
 - 1 : 3
 - 1 : 4
 - 1 : 6
- In practice, regenerative braking is used when
 - quick motor reversal is desired
 - load has overhauling characteristics
 - controlling elevators, rolling mills and printing presses etc.
 - other methods can not be used.
- Statement 1.** A direct-on-line (DOL) starter is used to start a small d.c. motor because
Statement 2. it limits initial current drawn by the armature circuit.
 - both statement 1 and 2 are incorrect
 - both statement 1 and 2 are correct
 - statement 1 is correct but 2 is wrong
 - statement 2 is correct but 1 is wrong
- Ward-Leonard system of speed control is **NOT** recommended for
 - wide speed range
 - constant-speed operation
 - frequent motor reversals
 - very low speeds
- Thyristor chopper circuits are employed for
 - lowering the level of a d.c. voltage
 - rectifying the a.c. voltage
 - frequency conversion
 - providing commutation circuitry

12. An inverter circuit is employed to convert
(a) a.c. voltage into d.c. voltage
(b) d.c. voltage into a.c. voltage
(c) high frequency into low frequency
(d) low frequency into high frequency
13. The phase-control rectifiers used for speed of d.c. motors convert fixed a.c. supply voltage into
(a) variable d.c. supply voltage
(b) variable a.c. supply voltage
(c) full-rectified a.c. voltage
(d) half-rectified a.c. voltage
14. If some of the switching devices in a convertor are controlled devices and some are diodes, the convertor is called
(a) full convertor (b) semiconvertor
(c) solid-state chopper
(d) d.c. convertor
15. A solid-state chopper converts a fixed-voltage d.c. supply into a
(a) variable-voltage a.c. supply
(b) variable-voltage d.c. supply
(c) higher-voltage d.c. supply
(d) lower-voltage a.c. supply
16. The d.c. motor terminal voltage supplied by a solid-state chopper for speed control purposes varies.....with the duty ratio of the chopper
(a) inversely (b) indirectly
(c) linearly (d) parabolically

ANSWERS

1. d 2. b 3. d 4. d 5. c 6. b 7. c 8. b 9. c 10. b 11. a
12. b 13. a 14. b 15. b 16. c